FALI: Mechanistic Decomposition of Conditional Attention-Head Importance

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Abstract

Component importance in transformer circuits can change after another component is perturbed, but detecting such conditional importance does not explain why the change occurred. We introduce FALI (Factorized Attention-Level Intervention), an output-grounded counterfactual framework that decomposes the cross-state change in a selected attention head's causal usefulness into target scores, rival scores, other attention scores, value content, and surrounding-network context. Standard Owen attribution allocates interactions while reconciling the fine score decomposition with a coarse score/value/context game. Controlled synthetic systems and a randomly initialized transformer validate factor recovery, endpoint identities, intervention isolation, and coarse–fine consistency. In GPT-2 Small on indirect object identification, a genuinely recruited head exhibits a positive target-score contribution and a negative rival-score contribution, while another head shows similar attention adaptation without positive recruitment. We then preregister transfer of this fine-score mechanism to Pythia-160M induction. Conditional recruitment appears again, but the target and rival contributions reverse sign. Thus, across the two studied settings, a common functional response does not imply a common fine-grained mechanism. FALI provides a mechanism-discrimination layer for selected conditionally important attention heads under explicit intervention semantics.

Keywords: mechanistic interpretability, causal intervention, transformer circuits, attention heads, conditional importance

1 Introduction

A component's causal importance in a transformer need not be stable under intervention. Removing one component can change the contribution of others: downstream layers can compensate for ablated computation, individual attention heads can change functional role depending on which other heads are available, and components that appear unimportant in the intact model can become important after a primary circuit is removed (McGrath et al. 2023; McDougall et al. 2023; Rushing and Nanda 2024; Nam et al. 2025; Gong et al. 2026). This conditionality is not merely a complication of terminology. Conditional co-ablation can recover backup components that first-order ablation scores miss, while causal-mediation analyses of activation patching show that a component's measured effect can contain substantial interactions with the state of the rest of the model (Gong et al. 2026; Vaidyanathan et al. 2026).

These results sharpen a question that localization alone does not answer. Suppose an attention head is known to become more or less useful after another circuit component is perturbed. *What changed in and around that head that accounts for the shift in its causal usefulness?* A changed ablation effect does not distinguish among several possibilities. The head may route attention differently, the information carried by its value vectors may change, or an otherwise similar head output may become more or less useful because the surrounding network has changed. Even within changed routing, a task-relevant attention edge can strengthen because its own score increases, because rival scores decrease, or because score changes elsewhere alter the head's computation. Existing work provides important pieces of this picture—including conditional importance, interaction-aware attribution, and causal separation of query–key routing from value-mediated computation—but these pieces target different estimands (Gong et al. 2026; Vaidyanathan et al. 2026; Wang et al. 2026).

We introduce **FALI (Factorized Attention-Level Intervention)**, a hypothesis-driven framework for resolving such conditional changes in a selected attention head. FALI begins with two model states: an intact state A and a perturbed state B. For a candidate head whose task-relevant target edge or edge set has already been specified, FALI factorizes the cross-state change into five intervention variables: target scores T, rival scores R, other structurally valid attention scores O, value content V, and surrounding-network context C. The first three factors jointly determine the candidate's attention pattern; V determines the content available to be transported; and C switches the surrounding computation between the intact and perturbed model states. For each of the 2^5 hybrid states, FALI defines the candidate's utility as its output-grounded contribution to the task metric: the metric with the hybrid candidate output injected minus the metric with that candidate zeroed in the same surrounding context. Thus the object being explained is not an attention probability, but the candidate head's change in causal usefulness.

Interactions among these factors make a purely sequential attribution arbitrary. We therefore use the standard Owen value with the prospectively fixed coalition structure $\{\text{SCORE} = \{T, R, O\}, \text{VALUE} = \{V\}, \text{CONTEXT} = \{C\}\}$ (Owen 1977). This gives a fine decomposition into $\phi_T, \phi_R, \phi_O, \phi_V, \phi_C$ while preserving a coarse score/value/context decomposition: $\phi_T + \phi_R + \phi_O$ equals the SCORE contribution in the quotient game. We make no claim of mathematical novelty for the Owen

value. The methodological object is the counterfactual intervention game and the architecture-aligned factorization to which it is applied.

FALI is deliberately downstream of circuit discovery. Path and activation patching localize components or paths that affect behavior (Goldowsky-Dill et al. 2023; Zhang and Nanda 2024; Heimersheim and Nanda 2024); conditional methods such as CoAx ask which components become important after a primary set is removed (Gong et al. 2026); interaction analyses establish that measured causal effects can depend on the rest of the network (Vaidyanathan et al. 2026). FALI instead takes a selected conditional change as its explanandum and asks which changes in the candidate's score computation, value content, or surrounding context account for it under an explicit intervention model. To our knowledge, FALI is the first framework to decompose the intervention-induced change in an attention head's output-grounded causal usefulness into architecture-aligned score, value, and surrounding-context contributions while resolving the score contribution into target, rival, and other-score channels. This claim is narrower than separating routing from value content itself, which is established in transformer-circuit analyses and recent causal decompositions (Elhage et al. 2021; Franco and Crovella 2025; Wang et al. 2026).

We validate the intervention semantics before using FALI on pretrained models. Deterministic synthetic attention systems isolate each of T, R, O, V, C in turn and test a known cross-factor interaction; a small randomly initialized transformer then checks score-to-pattern reconstruction, value-to-head-output reconstruction, endpoint identities, batch-specific score partitions, absence of live-candidate leakage, and agreement between the fine Owen decomposition and its coarse quotient game. These tests are intended to establish that the implemented decomposition identifies the factor manipulated in controlled systems, not that the chosen factorization is the unique natural decomposition of transformer computation.

We then study two substantively different settings. In GPT-2 Small on indirect object identification (IOI) (Wang et al. 2023), corrected functional analysis first shows why attention adaptation cannot be used as a proxy for recruitment: L10H2 becomes more causally useful after the canonical primary Name Movers are removed, whereas L10H10 increases target attention strongly while its total causal usefulness decreases. On broad canonical IOI templates, FALI assigns L10H2 a positive target-score contribution and a negative rival-score contribution; the same target-positive/rival-negative fine-score pattern also appears for L10H10 despite its negative total conditional gain. Tested matched-mean and matched-resample intervention variants preserve the preregistered fine-channel result, while a broader preregistered robustness gate fails because its zero-damage reference criterion is not satisfied; we report both outcomes.

Finally, we test a preregistered transfer of the GPT-2 fine-score mechanism to a different model family and task: Pythia-160M induction, using an established induction setting (Olsson et al. 2022). Conditional recruitment appears again, but the fine-score signs reverse: the target contribution is negative and the rival contribution positive. The transfer hypothesis therefore fails. Across the two studied settings, conditional recruitment is shared while the preregistered fine-score mechanism is not. This negative transfer result is central to the paper: it demonstrates the value of distinguishing

a common functional label from the mechanism that produces it, rather than treating recruitment itself as evidence for a single internal explanation.

Our contributions are threefold:

1. We define an output-grounded counterfactual intervention game for explaining the cross-state change in a selected attention head's causal usefulness.
2. We factor that change into score, value, and surrounding-context contributions, with target/rival/other-score resolution inside SCORE and coarse–fine consistency via standard Owen attribution.
3. We show that the same high-level conditional-recruitment phenomenon can conceal different fine mechanisms: GPT-2 IOI and Pythia induction exhibit opposite target/rival score signs under the specified FALI games.

2 Related Work

2.1 Self-repair and conditional component importance

Transformer circuits can adapt when components are ablated. The IOI circuit analysis of Wang et al. (2023) identified primary and backup Name Mover heads in GPT-2 Small, providing an early concrete setting in which several attention heads participate in a behavior and some become especially relevant under intervention. Subsequent work established self-repair more broadly. The Hydra effect describes compensatory changes in downstream language-model computation after ablation (McGrath et al. 2023). Copy-suppression analysis showed that self-repair can arise through specific functional mechanisms rather than a single generic backup process (McDougall et al. 2023). Rushing and Nanda (2024) further found self-repair across model families and the full training distribution, with substantial prompt-level variability and contributions from mechanisms including LayerNorm scaling and anti-erasure.

FALI does not claim the discovery of self-repair or heterogeneous compensation. More directly, it also does not introduce the idea of conditional component importance. Gong et al. (2026) formalize Conditional Co-Ablation (CoAx), which scores how much a remaining unit's ablation effect grows once a primary set has been removed. Their experiments show that this conditional score can recover backup heads missed by first-order rankings and transfers from GPT-2 IOI to induction settings across multiple models. This establishes both the empirical relevance and the practical utility of asking how component importance changes under intervention.

The distinction is therefore one of explanatory stage. Conditional ablation answers whether, or to what extent, a component becomes more important in a perturbed circuit. FALI assumes such a candidate has already been selected and asks which changes in the candidate's attention-score computation, value content, and surrounding context account for that cross-state change. In this sense CoAx and FALI are complementary: the former can identify *which* components warrant conditional analysis, whereas FALI provides a structured mechanism-resolution experiment for a chosen attention head.

2.2 Interaction-aware causal attribution

Activation patching and related interchange interventions are standard tools for testing whether internal activations causally contribute to model behavior. Path patching refines this idea to hypotheses about particular computational paths (Goldowsky-Dill et al. 2023); systematic studies have also shown that patching conclusions depend on choices such as corruption, metric, and intervention design (Zhang and Nanda 2024; Heimersheim and Nanda 2024). These methods are primarily localization or causal-testing tools: they identify components, activations, or paths whose replacement changes an outcome. FALI uses the same broad experimental tradition but addresses a different estimand—the change in one selected head's output-grounded causal contribution between two circuit states.

A particularly close conceptual neighbor is Vaidyanathan et al. (2026). Re-deriving activation patching through causal mediation analysis, they show that a component's natural indirect effect can include interaction terms that measure how its effect depends on the state of other model components. In GPT-2 IOI these interactions can hide conditionally important components, inflate others, and contribute to instability in component rankings. This result rules out any novelty claim that FALI is the first method to observe context-dependent causal effects. Instead, it motivates the next question: once a conditional or interaction-dependent effect has been identified, can we resolve that change into architecture-level factors of the selected attention head?

Causal Head Gating provides a complementary head-level perspective. Nam et al. (2025) learn gates over attention heads, classify their roles as facilitating, interfering, or irrelevant, and find that head roles can depend on interactions with other heads. FALI is neither a learned gating method nor a global head-role taxonomy. It fixes one candidate head and two model states, then intervenes on explicitly defined internal factors of that head and the surrounding model. Likewise, we do not claim that FALI is generally more faithful than patching: intervention-based interpretability results can be sensitive to methodology, as demonstrated by circuit-faithfulness studies (Miller et al. 2024). Our robustness experiments therefore support only the intervention variants actually tested.

These interaction-aware results also inform our interpretation of the context factor C. Under FALI, C is not a localized "downstream sensitivity" variable. It denotes the surrounding model computation that differs between intact and perturbed contexts outside the explicitly injected candidate score/value state. It can therefore aggregate changes in other heads, MLPs, residual representations, normalization, and upstream or downstream computation. A large ϕ_C is evidence about the specified counterfactual game, not identification of a unique external mechanism.

2.3 Decomposing attention computation

The separation between attention routing and transported content is foundational rather than novel to FALI. The transformer-circuits framework distinguishes the query–key (QK) circuit, which determines attention scores and patterns, from the output/value (OV) circuit, which determines what information is written when a source position is attended to (Elhage et al. 2021). Later causal work makes increasingly

fine distinctions inside attention computation. Franco and Crovella (2025) identify low-dimensional communication signals with a causal relationship to attention patterns, while Wang et al. (2026) causally separate Query–Key routing from Value updates when explaining how few-shot examples shape function vectors. Recent factorial intervention work likewise shows that attention allocation and token content can be experimentally separated yet strongly coupled in a different robustness setting (Zhu et al. 2026).

FALI builds on these distinctions but changes the object being decomposed. Its dependent quantity is not the attention pattern itself, nor the quality of a function vector, but the *cross-state change in the selected head's downstream causal usefulness*. SCORE and VALUE are therefore accompanied by an explicit CONTEXT factor. Within SCORE, FALI further partitions structurally valid score coordinates into task-target scores T, rival scores R in the same task-relevant query row, and all other valid scores O. This target/rival/other partition is hypothesis-driven: it requires a task-relevant target edge or edge set to be specified in advance, and is not an automatic circuit-discovery procedure.

Because score, value, and context can interact non-additively, FALI uses the standard Owen value for a game with a priori coalition structure (Owen 1977). The coalition structure is fixed as SCORE=$\{T, R, O\}$, VALUE=$\{V\}$, and CONTEXT=$\{C\}$. Owen attribution is established game-theoretic machinery; our use of it is motivated by the requirement that the fine score contributions reconcile with the coarse SCORE contribution rather than by a claim of a new attribution theorem. The resulting values should also be read relative to the intervention semantics. Hybrid internal states need not be states the unmodified model would naturally visit, and causal-intervention research has documented that representation-level interventions can move models away from their natural internal distributions (Grant et al. 2025). We therefore treat FALI as a controlled counterfactual decomposition, not as evidence that the model implements five independent natural pathways.

Finally, our emphasis on explicit validation reflects a broader evaluation problem in mechanistic interpretability. Benchmarks such as MIB prioritize precise recovery of causal pathways or variables rather than plausible narrative alone (Mueller et al. 2025). FALI's synthetic single-factor systems and implementation invariants serve the narrower purpose of checking that the intervention pipeline recovers known manipulated factors before empirical use. They do not remove the substantive limitations of target selection, coalition choice, or intervention dependence, which we address explicitly in Section 9.

3 Problem Formulation

3.1 Conditional causal usefulness

Let f be a transformer evaluated by a scalar task metric m. We consider two executions of the same task instance: an *intact* state A and a *perturbed* state B, obtained by applying a prespecified intervention to a primary circuit component or set of components. Let h denote a candidate attention head whose change in functional importance we want to explain. The task metric can differ across applications—for example, an

IO-minus-subject logit difference in IOI or an induction-specific logit score—but it is fixed within an analysis.

For context $s \in \{A, B\}$, let m_s^h be the task metric with the candidate head operating normally and let m_s^0 be the metric when that candidate is zeroed while all other aspects of context s are held fixed. We define the candidate's output-grounded causal usefulness in context s as

$$E_s(h) = m_s^h - m_s^0. \tag{1}$$

The intact and conditional effects are therefore $E_A(h)$ and $E_B(h)$, respectively. The cross-state change we seek to explain is

$$\Delta_h = E_B(h) - E_A(h). \tag{2}$$

In our experiments this quantity is called *conditional gain*. Positive Δ_h means that the candidate's contribution becomes more positive after the primary intervention; it does not by itself imply that the candidate is supportive in the perturbed state. We reserve *positive functional recruitment* for cases in which both $E_B(h) > 0$ and $\Delta_h > 0$. FALI itself is not restricted to positive recruitment: it decomposes positive or negative changes in causal usefulness.

This definition deliberately evaluates the candidate against a zero-candidate baseline in the *same* surrounding context. Consequently, a difference between $E_A(h)$ and $E_B(h)$ cannot be produced merely by a shift in the task metric that affects both the candidate-present and candidate-zero conditions equally. The estimand is the change in the candidate's marginal causal contribution under the specified intervention, not the raw performance difference between states A and B.

3.2 Mechanism-resolution objective

The problem begins after a candidate head and a task-relevant attention target have been chosen. They may come from prior circuit analysis, conditional co-ablation, patching, or a task-specific hypothesis. FALI does not search over heads, discover a target edge, or infer a circuit automatically. Its input is a selected head h, two model states A and B, a scalar task metric m, and a target query–key edge (or prespecified target edge set) whose score is meaningful for the task.

The objective is to account for Δ_h using changes in three architecture-level objects associated with the candidate: its pre-softmax attention scores, its value vectors, and the surrounding computation in which the resulting head output is used. The score tensor is then resolved more finely into the task-target score, rivals in the same query row, and scores at other query positions. This yields a controlled mechanism-discrimination question: under a fixed intervention algebra, which candidate-local or surrounding changes receive credit for the observed change in output-grounded usefulness? Figure 1 summarizes this distinction.

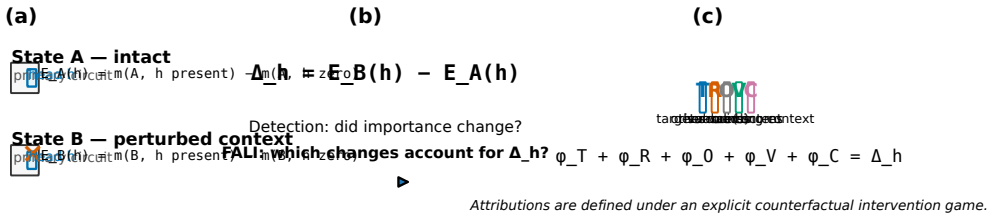


Fig. 1 FALI explains a conditional change rather than detecting it. A selected attention head has output-grounded causal usefulness $E_A(h)$ in an intact context and $E_B(h)$ after a prespecified circuit perturbation. Their difference Δ_h is the conditional gain to be explained. FALI constructs controlled hybrid interventions over target scores (T), rival scores (R), other valid scores (O), value content (V), and surrounding-network context (C), and allocates the endpoint change among these factors. The method is hypothesis-driven: the candidate head and task-relevant target are specified before FALI analysis.

4 Factorized Attention-Level Intervention

4.1 Architecture-aligned factors

For one candidate head and one task instance, let S^A and S^B denote its cached pre-softmax attention-score tensors in the intact and perturbed executions, and let V^A and V^B denote its corresponding value tensors. Let $q^\star$ be the task-relevant query position and $k^\star$ the target key position. We use an explicit structural validity mask

$$\mathcal{M} = \{(q, k) : q < \ell,\ k \le q\}, \tag{3}$$

for a right-padded causal sequence of valid length ℓ; padding and future positions are structural constants rather than intervention variables. The valid score coordinates are partitioned into

$$T = \{(q^\star, k^\star)\}, \tag{4}$$
$$R = \{(q^\star, k) \in \mathcal{M} : k \neq k^\star\}, \tag{5}$$
$$O = \mathcal{M} \setminus (T \cup R). \tag{6}$$

Thus T, R, and O are pairwise disjoint and cover every structurally valid score coordinate exactly once. When an application specifies multiple target edges, T can be generalized to that prespecified set and R defined as the remaining valid coordinates in the corresponding task-relevant query row(s); the experiments in this paper use one target coordinate per example.

FALI has five binary factors, each taking its value from state A (coded 0) or state B (coded 1):

- T: task-target attention score(s);
- R: rival scores in the same task-relevant query row;
- O: all other structurally valid scores of the candidate head;
- V: candidate value vectors at all valid source positions;
- C: surrounding-network context, intact versus perturbed.

The first three factors jointly determine the complete candidate attention pattern. V controls the content combined by that pattern. C controls the rest of the model execution outside the explicitly injected candidate state. In particular, C is not defined as a uniquely localized downstream mechanism; it can aggregate differences in other attention heads, MLPs, residual representations, normalization, and upstream or downstream computation.

4.2 Counterfactual intervention game

Let $(\tau, \rho, \omega, \nu, c) \in \{0,1\}^5$ denote one FALI cell. For a binary state index b, write $S^b = S^A$ if $b = 0$ and $S^b = S^B$ if $b = 1$, with the analogous convention for V^b. We construct a hybrid score tensor by taking each coordinate block from its selected endpoint,

$$S_{\tau\rho\omega} = \mathbf{1}_T \odot S^\tau + \mathbf{1}_R \odot S^\rho + \mathbf{1}_O \odot S^\omega, \tag{7}$$

with invalid coordinates fixed to the structural attention mask. We then recompute the attention pattern from the hybrid scores,

$$P_{\tau\rho\omega} = \text{masked_softmax}(S_{\tau\rho\omega}), \tag{8}$$

and reconstruct the candidate head output using the selected value tensor,

$$z_{\tau\rho\omega\nu}[q] = \sum_k P_{\tau\rho\omega}[q,k] V^\nu[k]. \tag{9}$$

The reconstructed z is injected for the candidate head at all valid query positions. This full-output overwrite is important: the live candidate computation is not allowed to leak into a hybrid cell.

Let $F_c[z_h \leftarrow z]$ denote a forward pass in surrounding context c, with $c = 0$ selecting state A and $c = 1$ selecting state B, and with the full output of candidate head h replaced by z. The FALI utility for a cell is

$$U(\tau, \rho, \omega, \nu, c) = m(F_c[z_h \leftarrow z_{\tau\rho\omega\nu}]) - m(F_c[z_h \leftarrow 0])\,. \tag{10}$$

The zero-candidate comparison is always evaluated in the same context c as the hybrid injection. FALI therefore contains $2^5 = 32$ counterfactual cells.

The two complete endpoints recover the effects in Eq. 1:

$$U(0,0,0,0,0) = E_A(h), \tag{11}$$
$$U(1,1,1,1,1) = E_B(h), \tag{12}$$

so that

$$\Delta_h = U(1,1,1,1,1) - U(0,0,0,0,0). \tag{13}$$

This endpoint identity is a required implementation invariant, not an approximation in the mathematical definition.

Most mixed cells need not correspond to an activation state encountered during an unmodified forward pass. FALI should therefore be read as an explicit

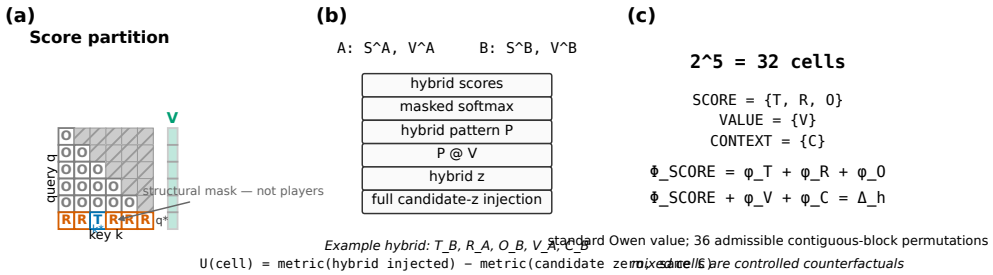


Fig. 2 FALI constructs an output-grounded 32-cell intervention game. Structurally valid attention scores are partitioned into the task-target coordinate(s) T, rivals R in the same query row, and all other valid scores O; V denotes the candidate value tensor and C the surrounding intact or perturbed computation. Each hybrid score tensor is passed through the model's masked softmax, combined with the selected values, and injected as the candidate's full head output. Utility is the task metric relative to a candidate-zero run in the same context. Standard Owen attribution under $\text{SCORE} = \{T, R, O\}$, $\text{VALUE} = \{V\}$, $\text{CONTEXT} = \{C\}$ allocates interactions while reconciling the fine score attributions with the coarse three-factor game.

counterfactual/interchange-intervention game. Its attributions are conditional on the chosen endpoint states, target partition, masking semantics, and intervention construction; they are not claims that the model naturally executes five separable causal pathways. Figure 2 depicts the 32-cell construction.

4.3 Standard Owen attribution and coarse–fine reconciliation

Let the player set be $N = \{T, R, O, V, C\}$ and associate each coalition $K \subseteq N$ with the cell in which factors in K take their B state and all remaining factors take their A state. This defines the characteristic function

$$v(K) = U(\mathbf{1}_K). \tag{14}$$

Because the score factors are subdivisions of one computational object, whereas value and context are separate coarse factors, we fix the coalition structure

$$\Pi = \{\text{SCORE} = \{T, R, O\},\ \text{VALUE} = \{V\},\ \text{CONTEXT} = \{C\}\}. \tag{15}$$

The partition membership is fixed; the block order is not. We use the standard Owen value, which averages marginal contributions over all permutations in which members of each block remain contiguous (Owen 1977). For this partition there are $3! \times 3! = 36$ admissible permutations.

For completeness, let $G \in \Pi$ be the block containing player i, let $M = |\Pi| = 3$, let $\mathcal{B} \subseteq \Pi \setminus \{G\}$ be a subset of predecessor blocks, and let $A_i \subseteq G \setminus \{i\}$ be a subset of predecessor players within G. Define $W_{\mathcal{B}} = \bigcup_{H \in \mathcal{B}} H$. The Owen value is

$$\phi_i = \sum_{\mathcal{B} \subseteq \Pi \setminus \{G\}} \frac{|\mathcal{B}|!\,(M - |\mathcal{B}| - 1)!}{M!} \sum_{A_i \subseteq G \setminus \{i\}} \frac{|A_i|!\,(|G| - |A_i| - 1)!}{|G|!} \tag{16}$$

$$\times \left[v(W_{\mathcal{B}} \cup A_i \cup \{i\}) - v(W_{\mathcal{B}} \cup A_i) \right]. \tag{17}$$

This is equivalent to averaging sequential marginal contributions over the 36 admissible permutations. We use independent subset-weight and permutation implementations in validation.

The first property needed for interpretation is efficiency:

$$\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = v(N) - v(\varnothing) = \Delta_h. \tag{18}$$

The attributions therefore allocate the entire endpoint change in causal usefulness, including interaction effects, under the specified game. They should not be interpreted as pure direct effects.

The second property gives the coarse–fine bridge. Define the quotient game on the three meta-players by

$$q(\mathcal{K}) = v\left(\bigcup_{G\in\mathcal{K}} G\right), \qquad \mathcal{K} \subseteq \Pi. \tag{19}$$

Standard Owen group consistency gives

$$\Phi_{\text{SCORE}} = \phi_T + \phi_R + \phi_O = \text{Shapley}_q(\text{SCORE}), \tag{20}$$

$$\phi_V = \text{Shapley}_q(\text{VALUE}), \tag{21}$$

$$\phi_C = \text{Shapley}_q(\text{CONTEXT}). \tag{22}$$

When the whole SCORE block switches together, its hybrid scores reproduce the complete A or B candidate pattern. The quotient SCORE/V/C game therefore matches the earlier pattern/value/context intervention game at the corresponding cells; this numerical bridge is tested explicitly in Section 5.2. The purpose of Owen attribution here is not mathematical novelty but consistency between a coarse score/value/context explanation and the finer $T/R/O$ explanation inside SCORE.

Finally, the historical two-factor attention-probability version of FALI is nested as a special case rather than identified with the present causal attributions. If the utility is changed from output-grounded usefulness to target attention probability, O, V, and C are held fixed, and only T and R vary, Eq. 17 reduces to the ordinary two-player Shapley decomposition of target-score versus competing-score changes. The present method changes both the intervention scope and, crucially, the quantity being explained.

5 Method Validation

Before applying FALI to pretrained models, we tested whether the implemented intervention game recovers known factors in controlled systems and whether its tensor-level interventions reproduce ordinary transformer computation at the two natural endpoints. These tests were preregistered as implementation and falsification checks rather than empirical evidence about any pretrained model. Figure 3 summarizes these validation checks.

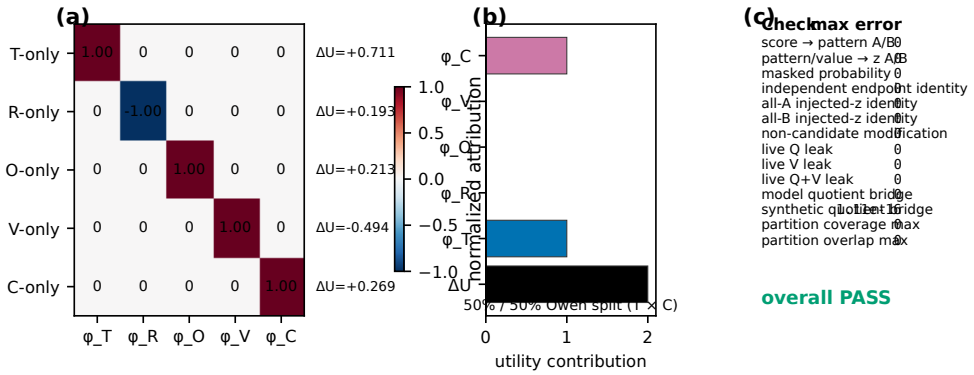


Fig. 3 FALI recovers controlled ground truth and satisfies implementation invariants. (a) In five deterministic attention systems where only one of T, R, O, V, or C differs between endpoints, the entire endpoint change is assigned to the manipulated factor up to floating-point error. Rows are normalized by their own absolute endpoint change for visual comparison; raw ΔU is annotated. (b) A constructed pure $T \times C$ interaction yields an equal Owen split between T and C, as analytically expected. (c) A randomly initialized two-layer TransformerLens model independently validates score-to-pattern and pattern/value-to-output reconstruction, endpoint identity, batch-safe partitions, candidate-isolation, live-computation leakage checks, and coarse–fine quotient reconciliation. These are implementation/ground-truth tests, not evidence that FALI's hybrid factors are unique natural pathways.

5.1 Analytical and synthetic ground truth

We first constructed deterministic attention systems in which the intact and perturbed states differed in exactly one FALI factor at a time. Each system included a causal score tensor, masked softmax, value vectors, attention output $z = PV$, and a context-specific downstream readout. We evaluated all 32 FALI cells and computed the standard Owen values under the frozen SCORE/VALUE/CONTEXT coalition structure.

In the five single-channel cases, only the intended factor received nonzero attribution. The target-score case produced $\Delta_h = 0.268952$ with $\phi_T = 0.268952$ and all other channels exactly zero at recorded precision; the rival-score case produced $\Delta_h = -0.494100$ with only ϕ_R nonzero. The corresponding O, V, and C constructions likewise assigned the complete endpoint change to the manipulated channel. Across these cases, the largest discrepancy between the intended contribution and the recovered attribution was 1.11×10^{-16}.

A separate construction tested a pure cross-factor interaction rather than an additive single-channel effect. We designed a $T \times C$ case with total endpoint change $\Delta_h \approx 2$ and no marginal effect from either factor in isolation. Standard Owen attribution split this interaction symmetrically: $\phi_T \approx 1$ and $\phi_C \approx 1$, with R, O, and V equal to zero. This test is important because an earlier fixed-block-order formulation would have assigned the interaction asymmetrically; the final method instead uses standard Owen attribution over all admissible block orders.

The synthetic systems also verified the endpoint and coarse–fine identities. All-A and all-B hybrid scores, patterns, and head outputs reconstructed their cached endpoints with zero recorded error. The endpoint difference matched the independently constructed conditional gain exactly. Collapsing T, R, O into SCORE and comparing

the resulting three-player SCORE/VALUE/CONTEXT game with the corresponding coarse factorial decomposition gave a maximum discrepancy of 1.11×10^{-16}.

5.2 Transformer implementation invariants

We next repeated the mechanical checks in a randomly initialized two-layer TransformerLens model rather than a hand-constructed attention system. The model had two attention heads per layer, $d_{\text{model}} = 32$, $d_{\text{head}} = 16$, $d_{\text{mlp}} = 64$, context length 8, and vocabulary size 48. A layer-0 head served as the primary perturbed component and a layer-1 head as the FALI candidate. The model was evaluated in FP32 on CPU with deterministic seed 4242; it was not pretrained and these results are not treated as evidence about learned transformer mechanisms.

The implementation first had to reproduce the model's own candidate computation. Applying masked softmax to the cached attention scores reconstructed both intact and perturbed attention patterns with zero recorded error, and multiplying those patterns by the cached value tensors reconstructed the candidate z tensors with zero recorded error. Masked coordinates received zero probability. Batch-specific partitions were also checked with different target keys across examples; every example had exactly one target coordinate, zero overlap among T, R, and O, and complete coverage of structurally valid score coordinates.

We then compared the two complete FALI endpoints against four independent ordinary forward passes: candidate present and candidate zeroed in the intact context, and candidate present and candidate zeroed in the perturbed context. The direct intact effect was 0.455046, the direct conditional effect was -0.085684, and their difference was -0.540729. The corresponding FALI endpoint cells reproduced all three quantities with zero recorded discrepancy. The injected all-A and all-B candidate outputs also matched the independently cached candidate outputs exactly.

Finally, we tested intervention isolation. Overwriting the candidate head left every non-candidate head unchanged at the injection site. Deliberately applying large perturbations to the candidate's live query computation, value computation, or both had no effect on the final metric once the full candidate z was overwritten; all three leakage errors were zero at recorded precision. The fine FALI game and its collapsed SCORE/VALUE/CONTEXT quotient also agreed exactly in this model-level test.

These checks establish a limited but necessary claim: the implemented FALI intervention pipeline behaves as specified in controlled systems, including factor isolation, interaction allocation, structural masking, endpoint recovery, and coarse–fine reconciliation. They do not establish that the chosen factorization is unique, that mixed hybrid states are on-manifold, or that a large attribution identifies a unique natural causal pathway. Those interpretation limits remain part of the method and are revisited in Section 9.

6 GPT-2 IOI: Conditional Recruitment and Fine-Score Decomposition

6.1 Experimental design

We first apply FALI to GPT-2 Small on indirect object identification (IOI), a setting with a well-characterized attention-head circuit and previously identified Name Mover and Backup Name Mover heads (Wang et al. 2023). The perturbed state B zero-ablates the three canonical primary Name Movers L9H6, L9H9, and L10H0 at their full head outputs. The intact state A leaves these heads unmodified. Our task metric is the final-position logit difference between the indirect-object (IO) token and the subject (S) token.

The confirmatory FALI analysis uses 15 low-level IOI templates and both BABA and ABBA prompt structures, giving 30 template–pattern strata. For each of four independently generated prompt cohorts (seeds 42, 1, 8, and 22), we sample 30 prompts per stratum, for 900 prompts per seed and 3600 prompts in total. The two primary candidate heads are L10H2 and L10H10. L9H7 and L9H0 serve as architectural controls because the primary intervention occurs at the same layer as those heads: under a full L9 `hook_z` intervention, their own cached score and value tensors are unchanged between A and B.

This stage is confirmatory specifically for the fine-grained score channels ϕ_T, ϕ_R, and ϕ_O. Coarser pattern/value/context quantities had already been analyzed on these canonical cohorts before the fine FALI hypotheses were frozen, so we do not treat those coarse quantities as held-out confirmation. The preregistered fine-channel hypotheses were $\phi_T > 0$ and $\phi_R < 0$ for each of L10H2 and L10H10; no directional hypothesis was made for ϕ_O.

Point estimates equally weight the 30 strata within each seed and then equally weight the four seed cohorts. Uncertainty is estimated with 10,000 deterministic hierarchical template-cluster bootstrap draws: low-level template clusters are sampled with replacement, their paired BABA/ABBA strata are retained, prompts are independently resampled within each seed–template–pattern cell, and the four seed means are then averaged equally. A preregistered strict non-overlap sensitivity excludes low-level template 0, which exactly overlaps a legacy discovery template, leaving 14 template clusters and 28 strata. Thus the main fine-channel conclusions are required to survive both broad-template sampling and removal of the known overlap.

6.2 Conditional recruitment is not attention increase

Before interpreting FALI channels, we re-establish the functional quantity being explained. In the broad canonical cohort, L10H2 has a positive conditional effect of 0.382 (95% hierarchical CI $[0.341, 0.424]$) and a positive conditional gain of 0.298 ($[0.263, 0.335]$). It therefore satisfies our definition of positive functional recruitment: after the primary Name Movers are removed, L10H2 remains beneficial and becomes more causally useful.

Target attention alone does not identify this functional response. L10H2 increases attention to the IO target by 0.279 ($[0.262, 0.296]$), but L10H10 shows an even larger

target-attention increase of 0.310 ([0.285, 0.335]) while its conditional gain is negative, −0.263 ([−0.302, −0.225]). On this broad canonical distribution, L10H10's conditional effect is also negative, −0.079 ([−0.111, −0.046]). Thus the larger visible attention adaptation occurs in a head that becomes less useful and, on average over these canonical prompts, harmful after the primary knockout.

This broad-template result also clarifies a distributional qualification from the earlier corrected functional screen. On the initial mixed two-template cohorts, L10H10 remained supportive after the knockout but was downregulated in relative usefulness; on the broader canonical cohort its average conditional effect crosses below zero. The stable conclusion across these analyses is narrower: a strong attention increase is not sufficient evidence of positive functional recruitment. We therefore treat the output-grounded conditional change, rather than attention change, as the FALI estimand.

6.3 Canonical FALI mechanism

FALI resolves the canonical cross-state changes into score, value, and surrounding-context contributions. For L10H2, the coarse SCORE contribution is positive ($\Phi_{\mathrm{SCORE}} = 0.445$) while the surrounding-context contribution is negative ($\phi_C = -0.146$); the value contribution is near zero ($\phi_V = -0.00035$). The fine score decomposition is strongly opposed: the target-score contribution is $\phi_T = 0.783$ (95% CI $[0.722, 0.838]$), while the rival-score contribution is $\phi_R = -0.341$ ($[-0.372, -0.311]$). The remaining-score contribution is small and positive ($\phi_O = 0.0039$). All four seed means have the preregistered signs, and all 30 template–pattern strata have positive ϕ_T and negative ϕ_R means.

L10H10 shows the same fine-score sign pattern despite its negative total conditional gain. Its target-score contribution is positive, $\phi_T = 0.329$ ($[0.300, 0.355]$), and its rival contribution negative, $\phi_R = -0.126$ ($[-0.139, -0.113]$); $\phi_O = 0.0036$. At the coarse level, SCORE is positive (0.206) but is outweighed by a much larger negative surrounding-context contribution ($\phi_C = -0.471$), yielding the negative endpoint change. This distinction is useful mechanistically: a candidate can exhibit a favorable-looking score adaptation under the specified game without becoming a functionally recruited backup.

The preregistered EF-4 fine-score gate therefore passes for both heads. Each of the four directional hypotheses is confirmed in all four seeds and all 30 canonical strata. The stricter template-0-excluded analysis also retains the same directions: L10H2 has $\phi_T = 0.776$ with CI $[0.713, 0.836]$ and $\phi_R = -0.337$ with CI $[-0.369, -0.306]$; L10H10 has $\phi_T = 0.325$ with CI $[0.294, 0.353]$ and $\phi_R = -0.123$ with CI $[-0.136, -0.111]$. The two same-layer controls behave as architectural invariants: across the canonical runs, L9H7 and L9H0 have zero recorded T, R, O, and V attributions, with their endpoint changes allocated to C up to numerical residual 1.11×10^{-16}.

These results do not imply that positive target and negative rival contributions constitute a universal recruitment mechanism. First, L10H10 displays the same fine-score signs without positive recruitment. Second, the FALI values allocate interactions under the specified intervention game rather than identify pure direct effects. We therefore treat the GPT-2 result as a well-supported mechanism pattern in this setting and test its transfer separately in Section 7.

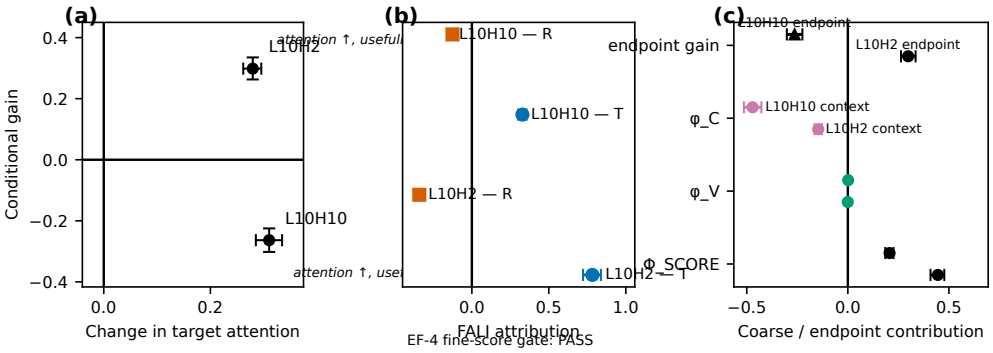


Fig. 4 Target-attention adaptation and positive functional recruitment are distinct in GPT-2 IOI. (a) L10H2 and L10H10 both increase attention to the IO target after the canonical primary Name Movers are ablated, but their conditional gains have opposite signs: L10H2 becomes more causally useful whereas L10H10 becomes less useful on the broad canonical distribution. (b) Both heads nevertheless show positive target-score and negative rival-score FALI contributions; the preregistered EF-4 fine-score gate passes. (c) Coarse attribution clarifies the distinction: L10H10's positive SCORE contribution is outweighed by a substantially negative surrounding-context contribution, producing a negative endpoint change. Points are exact balanced means; intervals are hierarchical 95% bootstrap CIs over the canonical template design.

6.4 Intervention robustness

We next asked whether the four preregistered GPT-2 fine-channel signs depend on the primary-damage intervention. In addition to the canonical zero-ablation analysis, EF-5 evaluated matched-mean and matched-resample primary-damage schemes on the same broad template design. Under matched-mean damage, all four channels retained their original directions with hierarchical CIs excluding zero: L10H2 had $\phi_T = 0.440$ ($[0.408, 0.470]$) and $\phi_R = -0.166$ ($[-0.178, -0.152]$), while L10H10 had $\phi_T = 0.188$ ($[0.173, 0.202]$) and $\phi_R = -0.0816$ ($[-0.0885, -0.0744]$). Under matched-resample damage, the same signs remained: L10H2 $\phi_T = 0.265$ ($[0.245, 0.285]$), $\phi_R = -0.0474$ ($[-0.0572, -0.0380]$); L10H10 $\phi_T = 0.0858$ ($[0.0776, 0.0941]$), $\phi_R = -0.0150$ ($[-0.0189, -0.0114]$). Each tested alternative was classified ROBUST by the preregistered channel criteria, with the predicted direction in all 30 strata.

However, the preregistered EF-5 experiment-level gate did *not* pass. Its reference-perturbation criterion required the zero-ablation damage-drop hierarchical confidence interval to lie entirely above zero; the observed zero-reference quantity was -0.319 with CI $[-0.396, -0.244]$, so that assumption failed and triggered the preregistered stop condition. We therefore make the bounded robustness claim supported by the data: the target-positive/rival-negative fine-channel signs persist under the matched-mean and matched-resample alternatives that were tested. We do not claim robustness to arbitrary intervention choices, and we report the failed overall EF-5 gate rather than treating the successful channel checks as a global pass.

7 Cross-Setting Test: Pythia Induction

7.1 Preregistered transfer design

The GPT-2 IOI results establish a stable target-positive/rival-negative fine-score pattern under the FALI game, but they do not establish that this organization transfers beyond that task and model. We therefore preregistered one cross-setting test using Pythia-160M on synthetic repeated-token induction. The transfer hypothesis was deliberately specific: if a later induction head became more causally useful after an earlier induction head was removed, we predicted the same fine-score directions observed in GPT-2, $\phi_T > 0$ and $\phi_R < 0$. Failure of either direction was to count against transfer rather than trigger selection of another model, task, or head pair.

The induction dataset contains a length-32 block of distinct tokens followed by a repeated prefix. For query offsets $j \in \{8, 16, 24\}$, the model receives $[x_0, \ldots, x_{31}, x_0, \ldots, x_j]$ and is evaluated on the log probability of x_{j+1} at the final query position $q = 32 + j$. A matched control replaces the first block with a disjoint token sequence while preserving the repeated suffix. The FALI target edge is fixed as (q, k) with $k = j + 1$, corresponding to the induction source position. The scalar task metric is the target-token log probability; induction advantage is the difference between repeated and control inputs.

Head selection was separated from FALI attribution. Discovery used seed 42 only, with 128 examples at each offset (384 total), and was forbidden from computing or inspecting any $\phi_T, \phi_R, \phi_O, \phi_V, \phi_C$, or Φ_{SCORE} quantity. Primary heads had to exhibit positive clean induction attention and a positive intact causal effect; candidate heads had to occur later in the network and show positive conditional effect and positive conditional gain after primary ablation. Among eligible ordered pairs, the preregistered rule selected the pair with the largest balanced conditional gain. Of 241 evaluated pairs, 238 satisfied the eligibility rules; this procedure selected L5H0 as the primary head and L8H2 as the candidate. On the discovery cohort, L8H2 had conditional gain 1.684 and conditional effect 1.804. The selected pair was then frozen before confirmation.

Confirmation used three independent prompt-generation cohorts (seeds 1, 8, and 22), again with 128 examples at each of the three offsets, giving 384 examples per seed and 1152 held-out examples in total. Observed quantities are balanced first across offsets within seed and then equally across seeds. Confidence intervals use 10,000 deterministic bootstrap draws with resampling inside each seed–offset cell; seeds themselves are not resampled. Five gates were frozen: induction behavior (G0), causal importance of the primary head (G1), positive conditional recruitment of L8H2 after L5H0 ablation (G2), transferred $\phi_T > 0$ (G3), and transferred $\phi_R < 0$ (G4). The experiment-level gate required all five to confirm.

7.2 Recruitment generalizes but the fine mechanism reverses

The held-out Pythia cohorts satisfy the behavioral and functional prerequisites. Repeated sequences yield an induction advantage of 14.017 log-probability units (95% CI $[12.280, 15.939]$). Ablating L5H0 produces a positive primary drop of 1.146 ($[0.887, 1.539]$), confirming that the selected primary head is causally useful for the

induction metric. More importantly, L8H2 is positively recruited after that intervention: its conditional effect is 0.621 ([0.456, 0.860]) and its conditional gain is 0.486 ([0.326, 0.722]). Both quantities are positive in every confirmation seed and at every offset. Thus G0, G1, and G2 confirm independently of the FALI fine-score hypotheses.

The preregistered fine-score mechanism does not transfer. Instead of the predicted positive target contribution, FALI gives

$$\phi_T = -0.144, \qquad 95\%\ \text{CI} = [-0.247, -0.084], \tag{23}$$

with a negative mean in all three confirmation seeds and all three offsets. Instead of the predicted negative rival contribution, FALI gives

$$\phi_R = +0.118, \qquad 95\%\ \text{CI} = [0.058, 0.219], \tag{24}$$

with a positive mean in every confirmation seed and offset. The reversal is therefore not a near-zero failure of detection. Relative to the preregistered absolute score-channel denominator 0.261807, $|\phi_T|$ and $|\phi_R|$ account for approximately 54.9% and 45.0%, respectively, so both channels also clear the prespecified magnitude threshold in absolute value while having the wrong signs for transfer.

The remaining fine-score channel is negligible on average ($\phi_O = -0.00027$, descriptive only). At the coarse level, the value contribution is small and negative ($\phi_V = -0.00210$), whereas the surrounding-context contribution is positive and large ($\phi_C = 0.514$). Together with ϕ_T and ϕ_R, these quantities sum to the observed conditional gain under Owen efficiency, up to the recorded numerical precision. We do not interpret the large context term as a uniquely localized external mechanism; as defined in Section 4.1, it is the contribution assigned to the surrounding computation under the specified intervention game.

Accordingly, the preregistered EF-6 gate is FAIL: G0, G1, and G2 confirm, while both mechanistic transfer gates G3 and G4 do not. No alternate head pair, model, threshold, or task was substituted after this outcome. This is a failed universality hypothesis, not a failed recruitment result. In both studied settings a later attention head becomes more causally useful after a primary component is perturbed, but the fine target/rival score organization is opposite: GPT-2 IOI has $\phi_T > 0$ and $\phi_R < 0$, whereas Pythia induction has $\phi_T < 0$ and $\phi_R > 0$. The result therefore argues against treating the high-level label of conditional recruitment as evidence for one shared fine-score mechanism. It does not establish that no common recruitment mechanism exists across transformers generally; the inference is restricted to the two model/task settings tested here. Figure 5 summarizes the preregistered transfer result.

8 Discussion

8.1 A common functional response need not imply a common mechanism

The central empirical result is a failed mechanism-transfer hypothesis. In GPT-2 Small on IOI, L10H2 becomes more causally useful after the canonical primary Name Movers

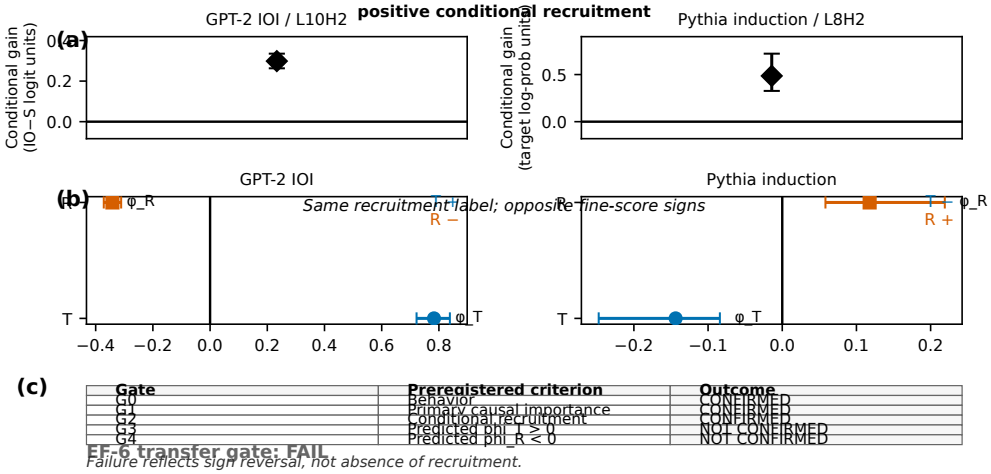


Fig. 5 Conditional recruitment occurs in both studied settings, but the preregistered fine-score mechanism reverses. (a) GPT-2 IOI L10H2 and Pythia-160M induction L8H2 both have positive conditional gain after their respective primary interventions; axes are kept task-specific because the metrics use different units. (b) Their FALI target/rival decompositions have opposite signs: GPT-2 is target-positive/rival-negative, whereas Pythia is target-negative/rival-positive. (c) Accordingly, the preregistered Pythia transfer test confirms induction behavior, primary-head importance, and conditional recruitment (G0–G2) but fails both sign-transfer gates (G3–G4), so EF-6 is a failed mechanism-transfer hypothesis rather than a failed recruitment result. Intervals are the preregistered 95% bootstrap CIs for each experiment.

are removed, and its FALI score decomposition is target-positive and rival-negative. In Pythia-160M induction, L8H2 also becomes more causally useful after an upstream induction head is removed, but the preregistered fine-score directions reverse: the target contribution is negative and the rival contribution positive. Conditional recruitment therefore occurs in both studied settings, while the specific GPT-2 fine-score organization does not transfer.

This distinction matters because the high-level label *recruitment* specifies a change in causal usefulness, not the internal route by which that change arises. Conditional-ablation methods can identify components whose importance increases after another component is removed (Gong et al. 2026); interaction analyses show that measured component effects can depend strongly on the state of the rest of the model (Vaidyanathan et al. 2026). Our results show why an additional mechanism-resolution step can be informative: two cases that satisfy the same functional criterion can receive opposite fine-grained score attributions under the same FALI intervention semantics.

The GPT-2 results also caution against treating a visually plausible attention adaptation as evidence of functional compensation. L10H10 increases target attention and has the same target-positive/rival-negative score pattern as L10H2, yet its total conditional gain is negative on the broad canonical distribution. Conversely, a positive or negative FALI channel should not be read in isolation as a statement that the head was recruited. Recruitment is defined by the output-grounded candidate effect; the

channel values explain the endpoint change under the specified game. This separation between observed attention, component usefulness, and mechanism attribution is central to the interpretation of FALI.

The cross-setting reversal is consistent with a broader picture in which transformer mechanisms can depend on prompts, components, and model state rather than forming a single invariant circuit description (Franco et al. 2026; Nam et al. 2025). Our evidence is much narrower: it concerns one GPT-2 IOI setting and one Pythia induction setting. It does not establish that fine mechanisms generally reverse across models, nor does it rule out shared mechanisms in other cases. It establishes that conditional recruitment itself is insufficient evidence for one shared target/rival mechanism.

8.2 Relationship to conditional attribution and patching

FALI is best viewed as downstream of localization and conditional-importance analysis. Activation and path patching test whether particular activations or paths causally matter for behavior (Goldowsky-Dill et al. 2023; Zhang and Nanda 2024; Heimersheim and Nanda 2024). Conditional co-ablation asks which components become more important after a primary set is perturbed (Gong et al. 2026). Interaction-aware mediation analysis shows that a component's causal effect can depend on the surrounding network (Vaidyanathan et al. 2026). FALI begins after such a candidate and task-relevant target have been specified and asks a different question: which changes in the selected attention head's score computation, value content, and surrounding context account for the cross-state change in its output-grounded usefulness under an explicit counterfactual intervention model?

This role also distinguishes FALI from generic QK/OV or routing/value decompositions. The separation between attention routing and transported content is established transformer-circuit structure (Elhage et al. 2021), and recent causal analyses resolve query–key routing and value-mediated computation in other settings (Franco and Crovella 2025; Wang et al. 2026). FALI uses these architectural objects to decompose a different estimand: the *change* in the selected head's downstream causal contribution across two circuit states. The additional context factor makes explicit that an otherwise similar candidate computation can acquire different usefulness when the rest of the model changes.

The use of standard Owen attribution should likewise be understood as allocation machinery rather than a claim about unique causal pathways. Interactions are distributed according to the prospectively fixed coalition structure, and the resulting ϕ_i values reconcile exactly with the coarse SCORE/VALUE/CONTEXT quotient game. This guarantees accounting consistency under the defined characteristic function; it does not make the decomposition invariant to every alternative intervention or coalition design. As with other intervention-based interpretability methods, conclusions can depend on the intervention semantics (Miller et al. 2024).

These distinctions suggest a practical workflow rather than a replacement for existing tools: first localize or identify a conditionally important component, then use FALI when the scientific question is why that selected attention head's causal usefulness changed. The method is therefore complementary to circuit discovery, patching, and conditional attribution rather than a substitute for them.

9 Limitations

Hypothesis-driven target selection.

FALI requires a selected attention head and a task-relevant target edge or edge set. In IOI and repeated-token induction, the target coordinates have clear task-specific interpretations. In less structured tasks, selecting a meaningful target may itself require substantial prior circuit analysis. FALI is consequently a mechanism-resolution framework, not an automatic circuit-discovery procedure.

Counterfactual and potentially off-manifold hybrids.

Mixed $T/R/O/V/C$ cells combine internal states that the unmodified model need not naturally realize. Their interpretation is therefore explicitly counterfactual: they answer what the specified model would do under the intervention construction, not which latent state the model would have visited naturally. Representation-level interventions can produce internal states that diverge from natural model trajectories (Grant et al. 2025). We therefore do not interpret FALI attributions as proof of five independent natural causal pathways.

Dependence on intervention and coalition semantics.

The decomposition depends on the intact and perturbed endpoints, the candidate-zero baseline, the structural score partition, and the fixed SCORE/VALUE/CONTEXT coalition structure. Standard Owen attribution allocates interactions consistently within that game, but another scientifically defensible intervention family or grouping could allocate interaction credit differently. Our robustness evidence is correspondingly bounded: the preregistered target-positive/rival-negative GPT-2 channels persisted under the matched-mean and matched-resample alternatives tested in EF-5, but the overall EF-5 experiment gate failed because its zero-damage reference criterion was not satisfied. We therefore do not claim robustness to arbitrary ablation choices.

The context factor is coarse.

C denotes the surrounding computation outside the explicitly injected candidate score/value state. It can aggregate changes in other attention heads, MLPs, residual representations, normalization, and upstream or downstream computations. A large ϕ_C therefore localizes explanatory credit only to the surrounding-context block of the specified game; it does not identify which external component or pathway generated that contribution. Further decomposition of C would require additional hypotheses and interventions.

Empirical scope.

The substantive evidence comprises two model/task settings: GPT-2 Small on IOI and Pythia-160M on repeated-token induction. The GPT-2 analysis spans broad canonical templates, multiple prompt-generation seeds, and tested ablation variants, while the Pythia test uses a separately frozen discovery/confirmation procedure. These experiments are sufficient to falsify the preregistered claim that the GPT-2 target/rival mechanism transfers unchanged to the tested Pythia induction setting, but they

are not sufficient to characterize the distribution of recruitment mechanisms across transformer architectures, scales, tasks, or training stages.

Attention-head-specific scope.

The current intervention algebra is designed around attention scores, values, and head outputs. Extending the same explanatory objective to MLPs, residual-stream features, or distributed multi-component mechanisms would require different architecture-aligned factors rather than mechanically reusing $T/R/O/V/C$. Thus the present results support FALI for selected attention-head mechanisms, not a universal component-attribution framework.

10 Conclusion

We introduced FALI, an output-grounded counterfactual framework for explaining how a selected attention head's causal usefulness changes after a circuit perturbation. FALI separates target, rival, and other attention-score changes from value content and surrounding-network context, and uses standard Owen attribution to reconcile the fine score decomposition with a coarse score/value/context explanation.

The experiments establish two complementary points. First, attention adaptation and functional recruitment are distinct: a head can increase task-target attention without becoming more causally useful. Second, the same high-level functional response can conceal different mechanisms. GPT-2 IOI and Pythia induction both contain positive conditional recruitment in the studied cases, but their preregistered target/rival score contributions have opposite signs. FALI therefore provides a mechanism-discrimination layer between identifying conditional importance and asserting an internal explanation for it.

The appropriate conclusion is not that transformers share one repair mechanism, nor that FALI recovers a unique natural causal decomposition. Rather, under explicit intervention semantics, FALI makes competing mechanistic explanations of conditional attention-head importance measurable and testable. In the settings studied here, that resolution is necessary to see that a common functional label does not imply a common fine-grained mechanism.

Statements and Declarations

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Data Availability

Code Availability

Author Contributions

Use of Generative AI

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